

MTH632 Final Exam ^{Fall} 2019

- ① Write $-3i^4$ in the form of $a+ib$. 2 Marks
- ② Find antiderivative of $\int_i^{i/2} e^{xz} dz$. 2 Marks
- ③ $Z_n = \frac{1}{n^3} - i$ solve it for convergence also find value of ϵ . 2 Marks
- ④ If $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ then show that it satisfies Laplacian eq. 3 Marks.
- ⑤ Evaluate $\bar{Z}_1 - \bar{Z}_2$ when $Z_1 = 5+i$ & $Z_2 = -8+3i$ 3 Marks.
- ⑥ If $\int_C f(z) dz$ & $f(z) = \frac{z+2}{z}$ & $z = 2e^{i\theta}$ 5 Marks.
- ⑦ $\int_0^{i+2i} \cos\left(\frac{z}{2}\right) dz = e + \frac{1}{e}$ 5 Marks
Prove it.
- ⑧ Find df and $df(V_p)$ if $f = (x^2-1)y + (y^2-1)x$ 5 Marks
- ⑨ $\hat{t} = \frac{x'(t)}{|x'(t)|}$ Find when $x(t) = a \cos(t) \hat{e}_1 + a \sin(t) \hat{e}_2 + b(t) \hat{e}_3$. 3 Marks

Mth 632

25/8/2019

- 1) If $u(x, y) = e^y \cos x$ then find $v(x, y)$.
 - 2) $z_n = \frac{1}{n^3} - i$ ($n=1, 2, 3, \dots$)
 $\lim_{n \rightarrow \infty} (z_n)$ then show that it is convergent or not also find the value of ε .
 - 3) Find singularity for function
 $f(z) = \frac{1}{z^2 + 2z + 2}$.
 - 4) Prove that $\overline{\cos(iz)} = \cos(\overline{iz})$.
 - 5) Show that $d/dz(\cos z) = -\sin z$.
 - 6) Frenet Serret formula
 $\bar{n} = T\bar{P} - K\bar{T}$.
 - 7) 2 questions were from Cauchy integral formula.
 - 8) 2 questions were to find either the function is analytic or not.
- MCQs were very easy.



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Mth 639 Final Paper

QNo1

Compute $\overline{z_1 - z_2}$, $z_1 = 1 + 4i$, $z_2 = (3 + 4i)$
(2 marks)

QNo2

If u, v and φ, ϕ the Poisson
 $fdg = f dg + df g$ (3 marks)

QNo3

Show $f'(z) = \overline{z}$ does not exist (2 marks)

QNo4

$u(x+iy)$ in domain d . Show
the harmonic conjugate of $v(x,y)$ (5 marks)

QNo5

Prove that $\frac{d}{dx} \cos x = -\sin x$ (3 marks)

Final MCQs

Mth 632 $-\frac{d}{dx} \int_0^x \cos(\frac{z}{2}) dz = 0 + \frac{1}{2} \times ⑤$

Q1 Prove $\int_0^x \cos(\frac{z}{2}) dz = 0 + \frac{1}{2} \times ⑤$

Q2 $x(t) = -a(\sin t)_1 + a(\cos t)_2 + b \cos$

Then solve $\frac{x(t)}{|x(t)|} \times ③$

Q3 $P(1,0) \quad V = (1,0,1)$

$f = 3x^2yz$

Then find $V_p[f] \times ⑤$

Q4 Check at what $f(z) = \frac{1}{z^2 + 2z + 2}$
 $\rightarrow$ Singular points $z = -1 \pm i$

~~Q5~~ ~~Prove~~ ~~$\arcsinh$~~

Q5 prove $\arcsinh z = \frac{1}{\sqrt{1+z^2}} \times ③$

Q6 If $\lim_{n \rightarrow \infty} |Z_n| = |Z|$ Then $\times ③$
 prove $\|Z_n - Z\| \leq |Z_n - Z|$
 and show that $\lim_{n \rightarrow \infty} |Z_n| = |Z|$

Q7 Maclaurin series of $\tanh$

Today's Paper (24.08.2019)

(Mth682) (final term)

Q:- Prove $d(fg) = gdf + f dg$

Q:- $f(x^2-1)y + (y^2+2)z$
find df and $V_p[f]$.

Q:- $\frac{d}{dz} (\sin z) = \cos z$

Q:- Show $\sum_{n=1}^{\infty} \overline{z_n} = \overline{S}$

Q $\overline{\cos(iz)} = \cos(iz)$

Q $f(z) = \frac{1}{z^2 + 2z + 2}$ find Si
find singularity

Feynman formula

$$\vec{r} = \vec{r}_p - \vec{r}_t$$

Let C denote the positive oriented boundary of the square whose sides lie along the lines $x = \pm 2$, $y = \pm 2$

Evaluate integral $\int_C \frac{\cosh z}{z^4} dz$ by using extended Cauchy integral formulae.

If $\lim_{n \rightarrow \infty} z_n = z$ then by using the inequality

$$||z_n| - |z|| \leq |z_n - z|$$

Show that $\lim_{n \rightarrow \infty} |z_n| = |z|$